📘 ENGINEERING MATHEMATICS · B.TECH 1ST YEAR

Multiple Integrals

Unit V — Complete Formula & Concept Revision Sheet

📐 DOUBLE INTEGRAL

$\iint f(x,y)\, dA$

📦 TRIPLE INTEGRAL

$\iiint f(x,y,z)\, dV$

🔄 JACOBIAN

$dA = |J|\, du\, dv$

🌐 SPHERICAL VOL.

$r^2 \sin\theta\, dr\, d\theta\, d\varphi$

Double Integrals · Triple Integrals · Change of Order

Jacobian & Transformation · Polar Coordinates · Cylindrical Coordinates

Spherical Coordinates · Area by Integration · Volume by Integration

Exam Tips & Flashcards

DESIGNED FOR LAST-DAY EXAM REVISION · COMPACT · EXAM-READY · MEMORY-FRIENDLY

📊 DOUBLE INTEGRALS

Integration over a 2D Region

Definition & Standard Form

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

STANDARD ITERATED FORM — INTEGRATE INNER VARIABLE FIRST

GEOMETRIC MEANING

- **Volume** under surface $z = f(x,y)$
- Over region **R** in the xy-plane
- If $f(x,y) = 1 \rightarrow$ gives **Area** of R
- Positive f $\Rightarrow$ volume above xy-plane

TYPES OF REGIONS

- **Type I (X-simple):** $a \le x \le b$, $g_1(x) \le y \le g_2(x)$
- **Type II (Y-simple):** $c \le y \le d$, $h_1(y) \le x \le h_2(y)$
- Choose type based on ease of limits

Fubini's Theorem

$$\int_a^b \int_c^d f(x,y)\, dy\, dx = \int_c^d \int_a^b f(x,y)\, dx\, dy$$

VALID WHEN F IS CONTINUOUS ON RECTANGLE [A,B] × [C,D]

Key Properties

Property	Formula
Linearity	$\iint[\alpha f + \beta g]\, dA = \alpha\iint f\, dA + \beta\iint g\, dA$
Domain Split	$\iint_R f\, dA = \iint_{R_1} f\, dA + \iint_{R_2} f\, dA$ (if $R = R_1 \cup R_2$)
Comparison	$f \le g$ on $R \Rightarrow \iint f\, dA \le \iint g\, dA$
Area	$\text{Area}(R) = \iint_R 1 \cdot dA$

📦 TRIPLE INTEGRALS

Integration over a 3D Region

Definition & Standard Form

$$\iiint_E f(x, y, z)\, dV = \int\int\int f(x, y, z)\, dz\, dy\, dx$$

ITERATED TRIPLE INTEGRAL — INTEGRATE INNERMOST VARIABLE FIRST

GEOMETRIC MEANING

- Integration over a 3D solid region E
- If $f(x,y,z) = 1 \rightarrow$ gives **Volume** of E
- If f = density $\rightarrow$ gives **Mass**
- Extends double integral to one more dimension

TYPICAL LIMITS SETUP

- Outer: x from a to b
- Middle: y from $g_1(x)$ to $g_2(x)$
- Inner: z from $h_1(x,y)$ to $h_2(x,y)$
- Always set up innermost limits first

$$\text{Volume} = \iiint_E dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} dz\, dy\, dx$$

VOLUME OF SOLID E USING TRIPLE INTEGRAL

CHANGE OF ORDER OF INTEGRATION

Swapping Integration Order

Concept

Rewriting ∬ by exchanging the order of dx and dy while keeping the same region. Useful when one order leads to unsolvable integrals.

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f \, dy \, dx \iff \int_c^d \int_{h_1(y)}^{h_2(y)} f \, dx \, dy$$

SAME REGION — DIFFERENT INTEGRATION ORDER

Why It's Useful

1. Avoids difficult or impossible inner integrals
2. Simplifies the region description
3. Reduces computation complexity
4. Required when ∫ is not elementarily expressible

Step-by-Step Method

1. **Write the region R** explicitly from the given limits (as inequalities in x and y)
2. **Sketch the region** R on the xy-plane — identify all bounding curves
3. **Find new limits** — express the region with y as outer variable (or x if changing to dy dx)
4. **Rewrite the integral** with new limits and changed order
5. **Evaluate** the new integral

⚠️ **COMMON MISTAKE**

Always re-derive limits by sketching the region — **never just swap limits blindly**. The limits change form when the order changes.

CHANGE OF VARIABLES & JACOBIAN

Coordinate Transformation

General Transformation Formula

$$\iint_R f(x, y)\, dA = \iint_S f(x(u,v), y(u,v)) \cdot |J| \, du \, dv$$

TRANSFORM FROM (X,Y) TO NEW VARIABLES (U,V)

The Jacobian

$$J = \partial(x,y)/\partial(u,v) = |\partial x/\partial u \ \ \partial x/\partial v| = (\partial x/\partial u)(\partial y/\partial v) - (\partial x/\partial v)(\partial y/\partial u) \ \ |\partial y/\partial u \ \ \partial y/\partial v|$$

JACOBIAN DETERMINANT — SCALING FACTOR FOR AREA TRANSFORMATION

📌 JACOBIAN RULES

- Always use **|J|** (absolute value)
- $J \neq 0$ ensures valid transformation
- For polar: $J = r$ (so $dA = r\,dr\,d\theta$)
- For cylindrical: $J = r$ ($dV = r\,dr\,d\theta\,dz$)
- For spherical: $J = r^2 \sin\theta$

📌 FOR TRIPLE INTEGRALS

- Transform $(x,y,z) \to (u,v,w)$
- J is a 3×3 determinant
- $dV = |J|\, du\, dv\, dw$
- Used for cylindrical & spherical conversions

$$\iiint_E f(x,y,z)\, dV = \iiint_{E'} f(x(u,v,w), y(u,v,w), z(u,v,w)) \cdot |J| \, du \, dv \, dw$$

CHANGE OF VARIABLES FOR TRIPLE INTEGRALS

📐 POLAR COORDINATES

Best for Circular/Symmetric Regions

Transformation

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$r^2 = x^2 + y^2,\ \tan\theta = y/x$$

POLAR ↔ CARTESIAN RELATIONS

$$dA = r\,dr\,d\theta$$

AREA ELEMENT IN POLAR COORDINATES (JACOBIAN = R)

Polar Double Integral

$$\iint_R f(x,y)\,dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\,dr\,d\theta$$

DOUBLE INTEGRAL IN POLAR FORM

WHEN TO USE POLAR

- Circular/annular regions
- Integrand has $x^2 + y^2$ terms
- Symmetry about origin
- Integrals like $\iint e^{-(x^2+y^2)}\,dA$

⚠️ DON'T FORGET

The extra factor **r** in $dA = r\,dr\,d\theta$ is the Jacobian. Forgetting it is the most common polar coordinate mistake.

📏 CYLINDRICAL COORDINATES

Polar + Vertical z — Best for Cylinders & Cones

Transformation

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

CYLINDRICAL COORDINATES (R, Θ, Z)

Volume Element

$$dV = r\,dr\,d\theta\,dz$$

JACOBIAN = R (SAME AS POLAR IN XY-PLANE)

$$\iiint_E f\,dV = \int_\alpha^\beta \int_{r_1}^{r_2} \int_{z_1(r,\theta)}^{z_2(r,\theta)} f(r\cos\theta, r\sin\theta, z) \cdot r\,dz\,dr\,d\theta$$

TRIPLE INTEGRAL IN CYLINDRICAL COORDINATES

🌐 SPHERICAL COORDINATES

Best for Spheres & Cones with Apex at Origin

Transformation

$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$
$$r^2 = x^2 + y^2 + z^2$$

SPHERICAL COORDINATES (R, Θ, Φ) — R≥0, 0≤Θ≤Π, 0≤Φ≤2Π

Volume Element

$$dV = r^2\sin\theta\,dr\,d\theta\,d\varphi$$

JACOBIAN = R² SINΘ — MOST IMPORTANT FORMULA!

WHEN TO USE SPHERICAL

- Spherical regions: $x^2+y^2+z^2 = a^2$
- Cones centered at origin
- Integrands with $r^2 = x^2+y^2+z^2$

📊 MASTER COMPARISON TABLE — ALL COORDINATE SYSTEMS

Feature	Cartesian (x,y,z)	Polar (r,θ)	Cylindrical (r,θ,z)	Spherical (r,θ,φ)
Dimensions	2D / 3D	2D	3D	3D
x =	x	$r\cos\theta$	$r\cos\theta$	$r\sin\theta\cos\varphi$
y =	y	$r\sin\theta$	$r\sin\theta$	$r\sin\theta\sin\varphi$
z =	z	–	z	$r\cos\theta$
Area Element	$dx\,dy$	$r\,dr\,d\theta$	–	–
Volume Element dV	$dx\,dy\,dz$	–	$r\,dr\,d\theta\,dz$	$r^2\sin\theta\,dr\,d\theta\,d\varphi$
Jacobian |J|	1	r	r	$r^2\sin\theta$
r^2 relation	$x^2+y^2+z^2$	x^2+y^2	x^2+y^2	$x^2+y^2+z^2$
Best Used For	Rectangles, boxes	Circles, disks, sectors	Cylinders, cones	Spheres, cones at origin

📉 AREA & VOLUME FORMULAS

Using Multiple Integrals

Quantity	Formula	Coordinate System	Notes
Area of Region R	$\iint_R dA$	Cartesian	$f(x,y) = 1$
Area in Polar	$\int\int r\,dr\,d\theta$	Polar	Include Jacobian r
Volume (Double)	$\iint_R f(x,y)\,dA$	Cartesian	$z = f(x,y)$ above R
Volume (Triple)	$\iiint_E dV$	Cartesian	$f(x,y,z) = 1$
Volume (Cylindrical)	$\int\int\int r\,dz\,dr\,d\theta$	Cylindrical	Best for r-symmetric solids
Volume (Spherical)	$\int\int\int r^2\sin\theta\,dr\,d\theta\,d\varphi$	Spherical	For sphere: $\int_0^{2\pi}\int_0^{\pi}\int_0^{a}$ ...
Volume of Sphere	$(4/3)\pi a^3$	Spherical	r from 0→a, θ: 0→π, φ: 0→2π

$$\text{Area} = \iint_R dA \text{ (Cartesian)}$$
$$\text{Area} = \int_\alpha^\beta\int_0^{r(\theta)} r\,dr\,d\theta \text{ (Polar)}$$

AREA USING DOUBLE INTEGRALS

$$\text{Vol} = \iint_R f(x,y)\,dA \text{ (Double)}$$
$$\text{Vol} = \iiint_E dV \text{ (Triple)}$$

VOLUME USING MULTIPLE INTEGRALS

QUICK RECALL FLASHCARDS

Cards 1–14 of 25 | Tap Q → Recall A

Q1: What is a double integral?

Integration of f(x,y) over a 2D region R; gives volume under surface z = f(x,y)

Q2: Standard form of double integral?

$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$

Q3: If f(x,y) = 1 in ∬ f dA, result is?

Area of the region R

Q4: What is Fubini's Theorem?

$\iint f\,dy\,dx = \iint f\,dx\,dy$ for continuous f on a rectangle

Q5: Area element in polar coordinates?

$dA = r\,dr\,d\theta$ (r is the Jacobian)

Q6: Polar coordinate relations?

$x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$

Q7: What is a triple integral?

$\iiint f(x,y,z)\,dV$ — integration over a 3D solid region E

Q8: If f = 1 in ∭ f dV, result is?

Volume of the solid region E

Q9: Volume element in cylindrical coordinates?

$dV = r\,dr\,d\theta\,dz$ (Jacobian = r)

Q10: Cylindrical coordinate relations?

$x = r\cos\theta$, $y = r\sin\theta$, $z = z$

Q11: Volume element in spherical coordinates?

$dV = r^2 \sin\theta\,dr\,d\theta\,d\varphi$ (Jacobian = $r^2 \sin\theta$)

Q12: Spherical coordinate relations?

$x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$

Q13: What is the Jacobian for a transformation (x,y) → (u,v)?

$J = (\partial x/\partial u)(\partial y/\partial v) - (\partial x/\partial v)(\partial y/\partial u)$; use |J| in integral

Q14: Why change order of integration?

To simplify/evaluate integral when current order is unsolvable or complex

QUICK RECALL FLASHCARDS (Continued)

Cards 15–25 of 25

Q15: Steps to change order of integration?

1. Write region R 2. Sketch 3. Find new limits 4. Rewrite integral 5. Evaluate

Q16: Change of variables formula?

$\iint f(x,y)\, dA = \iint f(x(u,v), y(u,v)) \cdot |J|\, du\, dv$

Q17: When to use polar coordinates?

Circular regions or integrands containing x^2+y^2 or $\sqrt{x^2+y^2}$

Q18: When to use cylindrical coordinates?

Solids with circular cross-sections: cylinders, cones, paraboloids

Q19: When to use spherical coordinates?

Spheres, hemispheres, cones with apex at origin

Q20: Volume of sphere using spherical coords?

$\int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2 \sin\theta\, dr\, d\theta\, d\varphi = (4/3)\pi a^3$

Q21: What is r^2 in spherical coordinates?

$r^2 = x^2 + y^2 + z^2$ (distance from origin)

Q22: What is r^2 in cylindrical/polar?

$r^2 = x^2 + y^2$ (distance from z-axis)

Q23: Area of circle r=a in polar?

$\int_0^{2\pi} \int_0^{a} r\, dr\, d\theta = \pi a^2$

Q24: What is a Type I region?

$a \le x \le b$, $g_1(x) \le y \le g_2(x)$ — integrate y first, then x

Q25: Gaussian integral (polar trick)?

$\int_{-\infty}^{\infty} e^{(-x^2)}\, dx = \sqrt{\pi}$ (proved via double integral + polar)

✅ Memory Trick for Jacobians

Polar → r | Cylindrical → r | Spherical → $r^2 \sin\theta$. Remember: going from 2D to 3D in spherical adds an extra factor of r and sinθ compared to cylindrical.

⚠️ **Θ VS Φ NOTATION WARNING**

Different textbooks use different conventions for spherical coords. Always check: in this sheet, θ is the polar angle (from z-axis, 0 to π) and φ is azimuthal (0 to 2π).

EXAM TIPS & COMMON MISTAKES

Read Before Every Exam

❌ Mistakes with Limits

WRONG LIMIT DIRECTION

Always ensure lower limit < upper limit. If lower > upper, factor out a negative sign.

BLIND LIMIT SWAP

When changing order, **never swap limits directly**. Sketch the region and re-derive limits from scratch.

VARIABLE LIMITS CONFUSION

In $\int_a^b \int_{g_1(x)}^{g_2(x)} f\, dy\, dx$ — inner limits can depend on the outer variable (x), but outer limits must be **constants**.

❌ Mistakes with Jacobian

FORGETTING JACOBIAN

In polar: **always write r** in dA = r dr dθ. Missing r is the #1 polar mistake.

WRONG JACOBIAN SIGN

Use **|J|** (absolute value). Negative determinant means reflected region — area is still positive.

WRONG SPHERICAL JACOBIAN

Spherical Jacobian is $r^2 \sin\theta$, NOT r^2. The sinθ factor is often missed. Double-check every time.

❌ Coordinate System Mistakes

WRONG COORDINATE CHOICE

Using Cartesian for circular region = messy. Always match coordinate system to region geometry.

Θ RANGE ERROR

For a full circle: θ goes 0 to 2π. For upper half: 0 to π. For first quadrant: 0 to π/2.

SPHERICAL Θ RANGE

In spherical: θ (polar angle from z-axis) goes 0 to π, NOT 0 to 2π. φ goes 0 to 2π.

✅ Pro Strategies

Situation	Strategy
Inner integral can't be solved	Change order of integration — sketch region first
Region is circular or has x^2+y^2	Switch to polar coordinates
Region is a cylinder/cone	Use cylindrical coordinates
Region is a sphere	Use spherical coordinates
Limits are complex curves	Change variables with Jacobian
Computing $\iint e^{-x^2-y^2}\, dA$	Use polar — becomes $\iint e^{-r^2}\, r\, dr\, d\theta$ (solvable)
Need volume of standard solid	Identify symmetry → choose matching coordinate system

✅ 5-Minute Pre-Exam Checklist

☑ Remember dA = r dr dθ (polar) | ☑ Remember dV = r dr dθ dz (cylindrical) | ☑ Remember $dV = r^2 \sin\theta\, dr\, d\theta\, d\varphi$ (spherical) | ☑ Use |J| not J | ☑ Sketch region before changing order | ☑ θ in spherical → 0 to π only

🚀 MASTER QUICK REVISION — ALL FORMULAS AT A GLANCE

No explanations — pure formulas only

📊 DOUBLE INTEGRALS

STANDARD FORM
$\int_a^b \int_{g_1}^{g_2} f(x,y)\, dy\, dx$

AREA OF REGION R
$\iint_R dA$

VOLUME (DOUBLE)
$\iint_R f(x,y)\, dA$

FUBINI'S THEOREM
$\int\int f\, dy\, dx = \int\int f\, dx\, dy$

📦 TRIPLE INTEGRALS

STANDARD FORM
$\int\int\int f(x,y,z)\, dz\, dy\, dx$

VOLUME (TRIPLE)
$\iiint_E dV$

🔄 CHANGE OF VARIABLES (JACOBIAN)

GENERAL RULE
$dA = |J|\, du\, dv$

JACOBIAN (2D)
$J = (\partial x/\partial u)(\partial y/\partial v) - (\partial x/\partial v)(\partial y/\partial u)$

TRANSFORMATION
$\iint f\, dA = \iint f \cdot |J|\, du\, dv$

TRIPLE CHANGE
$dV = |J|\, du\, dv\, dw$

📐 POLAR COORDINATES

RELATIONS
$x = r\cos\theta,\ y = r\sin\theta$

AREA ELEMENT
$dA = r\, dr\, d\theta$

R² RELATION
$r^2 = x^2 + y^2$

AREA OF CIRCLE R=A
$\int_0^{2\pi} \int_0^a r\, dr\, d\theta = \pi a^2$

📏 CYLINDRICAL COORDINATES

RELATIONS
$x = r\cos\theta,\ y = r\sin\theta,\ z = z$

VOLUME ELEMENT
$dV = r\, dr\, d\theta\, dz$

🌐 SPHERICAL COORDINATES

RELATIONS
$x=r\sin\theta\cos\varphi,\ y=r\sin\theta\sin\varphi,\ z=r\cos\theta$

VOLUME ELEMENT
$dV = r^2 \sin\theta\, dr\, d\theta\, d\varphi$

R² RELATION
$r^2 = x^2 + y^2 + z^2$

SPHERE VOLUME
$(4/3)\pi a^3$

🎯 APPLICATIONS OF MULTIPLE INTEGRALS

Key Use Cases in Engineering & Physics

Application	Formula	Variables Used
Area of plane region	$A = \iint_R dA$	x, y or r, θ
Volume of solid	$V = \iiint_E dV$	x,y,z or r,θ,z or r,θ,φ
Mass of lamina	$M = \iint_R \rho(x,y)\, dA$	ρ = surface density
Center of mass ($\bar{x}$)	$\bar{x} = (1/M) \iint x \cdot \rho\, dA$	Weighted average position
Moment of inertia (I)	$I = \iint r^2 \cdot \rho\, dA$	r = dist. from axis

📋 COMPLETE CONCEPT SUMMARY TABLE

Topic	Key Idea	Critical Formula
Double Integral	Volume under f(x,y) over 2D region	$\iint f(x,y)\, dy\, dx$
Triple Integral	Integral over 3D solid region	$\iiint f(x,y,z)\, dz\, dy\, dx$
Change of Order	Swap dx↔dy; re-derive limits by sketching	Sketch → New limits → Integrate
Jacobian (2D)	Scaling factor in variable change	$J = \partial(x,y)/\partial(u,v)$
Polar	For circular regions	$dA = r\, dr\, d\theta$
Cylindrical	For solids with circular cross-sections	$dV = r\, dr\, d\theta\, dz$
Spherical	For spheres and symmetric 3D regions	$dV = r^2 \sin\theta\, dr\, d\theta\, d\varphi$
Area	Set f = 1 in double integral	$A = \iint dA$
Volume	Use triple integral with f = 1	$V = \iiint dV$

✅ Golden Rules for Exams

1. Always sketch the region before integration.
2. Match coordinate system to the geometry.
3. Never forget the Jacobian (r or $r^2 \sin\theta$).
4. Outer limits → constants. Inner limits → functions of outer variable.
5. When stuck — try changing order or coordinates.

$$\iint f\, dA = \iint f \cdot |J|\, du\, dv$$

CORE TRANSFORMATION PRINCIPLE

$|J|$: Polar = r | Cylindrical = r | Spherical = $r^2 \sin\theta$

JACOBIANS TO MEMORIZE